
Web Programming

— Lecture - 1 [Introduction] —

Learning Outcomes

After completing this course you should be able to

- Explore the landscape of **frontend** and **backend** web technologies
- Design and develop web pages using **HTML**, **CSS**, and **Javascript**
- Explore and use frontend web frameworks
- Develop **web applications**
- Develop **REST APIs**
- **Secure** web applications and REST APIs

Learning Outcomes

After completing this course you should be able to

- **Test** Web Applications and REST APIs
- Explore different **software architectures** available for web applications
- **Deploy** web applications

Website Design and Development Involves

Content design

Appearance and Structure design

All the front and backend technologies that make the website function/work



Website Design and Development Involves Roles

Information Architect

Content Strategist

User Experience, Interaction, and User Interface designer

Information Architect

An **Information Architect** (also called an **Information Designer**) organizes the content logically and for ease of findability

- responsible for search functionality,
- site diagrams, and
- how the content and data are organized on the server

Content Strategist

A **Content Strategist** makes sure that every bit of text on a site, from long explanatory text down to the labels on buttons, supports the brand identity and marketing goals of the organization

User Experience, Interaction, and User Interface design

User Experience (UX) design

about all aspects of the user's interaction with the product: **how it is perceived, learned, and used**

includes the **visual design**, the **user interface**, the **quality** and **message of the content**, and the **overall site performance**

the user experience must be in line with the organization's brand and business goals

User Experience, Interaction, and User Interface design

Interaction Design (IxD)

make the site as **easy**, **efficient**, and **delightful** to use as possible

User Interface (UI) design

focused on the **functional organization of the page** as well as the specific tools (**buttons, links, menus**, and so on) that users use to navigate content or accomplish tasks

Outputs of UX, IxD, and UI Design Phase

User research and testing reports

site designs often begin with **user research**, including **interviews** and **observations**, in order to gain a better understanding of how the site can solve user problems or how it will be used

The approach of designing around the user's needs is referred to as **User-Centered Design (UCD)**

Outputs of UX, IxD, and UI Design Phase

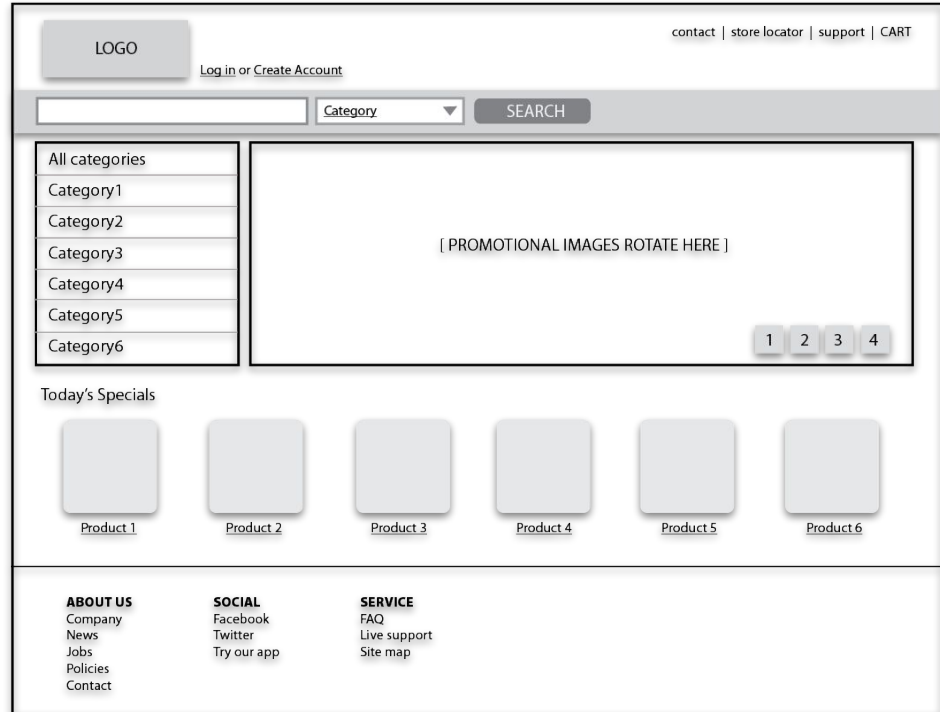
Wireframe Diagram

shows the structure of a web page using only outlines for each content type and widget

colors, fonts, and other visual identity elements are deliberately omitted so as not to distract from the structure of the page

Outputs of UX, IxD, and UI Design Phase

Wireframe Diagram



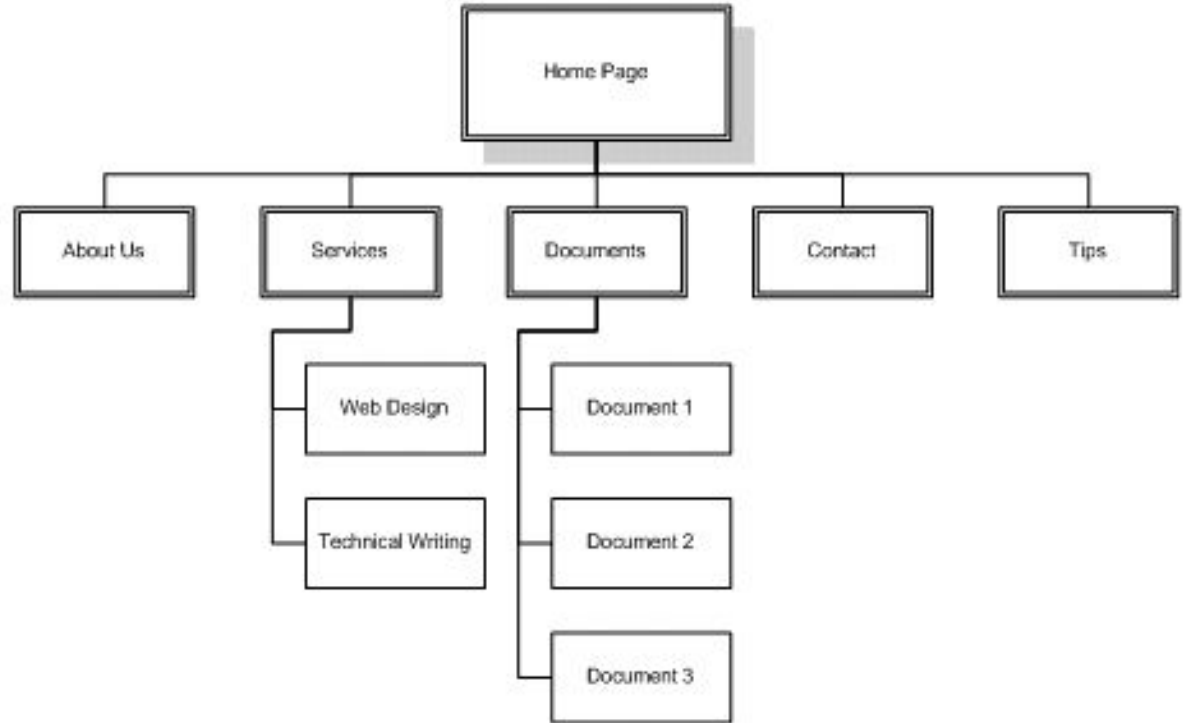
Outputs of UX, IxD, and UI Design Phase

Site Diagram

indicates the **structure** of the site as a whole and how individual pages **relate** to one another

Outputs of UX, IxD, and UI Design Phase

Site Diagram



Outputs of UX, IxD, and UI Design Phase

Storyboards

A **storyboard** traces the path through a site or application from the point of view of a typical user

The storyboard aims to **demonstrate the steps it takes to accomplish tasks**, outlines possible options, and also introduces some standard page types

Opportunities:

THE TWITTER OF TEAM SPACES: "TEAM UP"

Event:

SHARE A FILE

Storyboard



CAROL JUST UPLOADED A NEW REPORT — A REVISION. OTTO NOTICES, AND TAKES A QUICK PEEK TO SEE WHAT CHANGED.



A PREVIEW POPS UP, AND IT LOOKS LIKE CAROL MADE AN IMPORTANT CHANGE THAT MIGHT BE WORTH LOOKING AT — LATER.



BUT FOR NOW, OTTO NEEDS TO FINISH WHAT HE'S DOING. TEAM UP DOWNLOADS THE FULL FILE AND HOLDS ON TO IT FOR HIM TO CHECK OUT LATER.

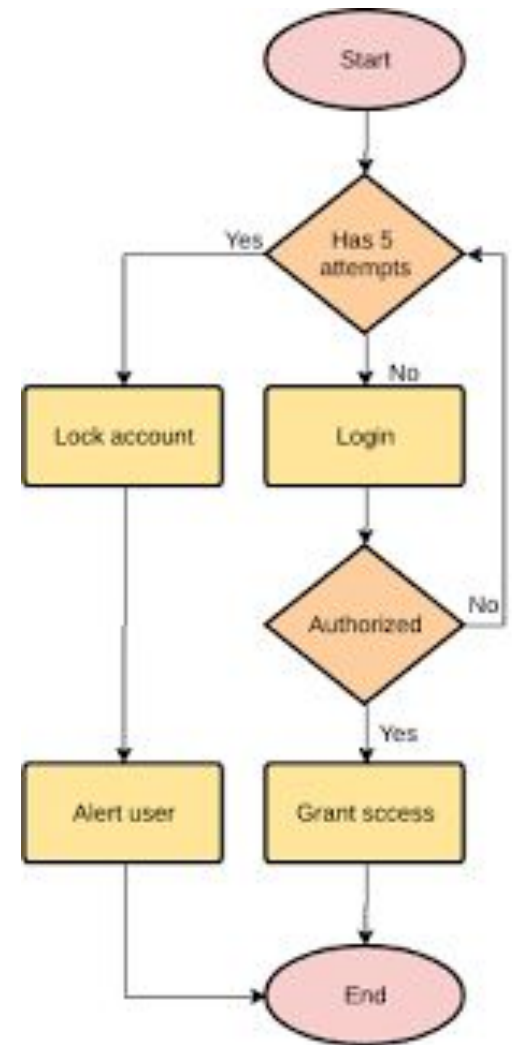
Outputs of UX, IxD, and UI Design Phase

User Flow Charts

A **user flow chart** is another method for showing how the parts of a site or application are connected, but it tends to focus on technical details rather than telling a story

Outputs of UX, IxD, and UI Design Phase

User Flow Charts



Outputs of UX, IxD, and UI Design Phase

Visual (graphic) design

A graphic designer creates the “**look and feel**” of the site—**logos, graphics, type, colors, layout**, and so on—to ensure that the site makes a good first impression and is consistent with the brand and message of the organization it represents

You can present a visual design to clients and stakeholders using tools such as Photoshop or styletil.es

Web Development

Development falls under two broad categories: **frontend** development and **backend** development

Frontend Development

Frontend refers to any aspect of the design process that appears in or relates directly to the browser

HTML, CSS, and JavaScript

Frontend Development

Authoring/markup (HTML)

is the process of preparing content for delivery on the web, or more specifically, **marking up the content with HTML tags** that describe its content and function such as headings, paragraphs, and lists

HTML (HyperText Markup Language) is the authoring language used to create web page documents

Frontend Development

Styling (CSS)

Cascading Style Sheets (CSS) describe how that content should look - presentation

fonts, colors, background images, line spacing, page layout, and so on, are all controlled with CSS

You can also add special effects and **basic animation** to your page

Frontend Development

JavaScript and DOM scripting (CSS)

JavaScript is a scripting language that **adds interactivity and behaviors** to web pages, including these

- Checking form entries for valid entries
- Swapping out styles for an element or an entire site
- Loading scrolling feeds with more content automatically

Frontend Development

JavaScript and DOM scripting (CSS)

DOM stands for **Document Object Model**, and it refers to the standardized list of **web page elements** that can be accessed and manipulated using JavaScript

Frontend Development

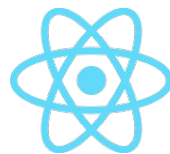
Frontend SPA Frameworks

React: A JavaScript library for building user interfaces

Angular: The modern web developer's platform

Vue.js: The Progressive JavaScript Framework

They are used for creating **Single Page Applications**



ANGULAR



Frontend Development

Frontend CSS Frameworks

Bootstrap (<https://getbootstrap.com/>)

Tailwind CSS (<https://tailwindcss.com/>)

BULMA (<https://bulma.io/>)

Materialize (<https://materializecss.com/>)

Backend development

Backend development focus on the server, including the applications and databases that run on it

Backend development

Technologies used for backend development

Programming Language	Web Frameworks
Java/Kotlin	Spring, Spring Boot, Quarkus, Micronaut
Javascript	Express.js, Nest.js, Node.js
Python	Django, Flask, Tornado
.NET	ASP.NET, ASP.NET Core
Golang	Gin, Beego

Backend development

Technologies used for backend data management

MySQL, Oracle, PostgreSQL, Microsoft SQL, MongoDB

Web Production Software

Coding tools

VS Code, Sublime Text, Atom, Bracket, Adobe Dreamweaver

User interface and layout tools

Affinity Designer, Adobe XD, **Figma**, UXPin

Web graphic creation tools

Adobe Photoshop, Adobe Illustrator, GIMP, Corel PaintShop Pro, Corel Draw, SumoPaint (sumopaint.com), **Pixlr** (pixlr.com)

Web Production Software

A variety of browsers

Chrome, Firefox, MS Edge, Internet Explorer 9–11, Safari, Opera

File management and transfer tools

Filezilla, Cyberduck, WinSCP, Transmit

How the web works

What is the difference between Web and Internet?

How the web works

The **internet** is an international network of connected computers

The purpose of connecting computers together is to share information

A standardized methods for transferring data or documents over a network are known as **protocols**

Mention some of the ways that you can share information over the internet

How the web works

There are many ways information can be passed between computers, including

- email** (POP3/IMAP/SMTP),

- file transfer** (FTP),

- secure shell** (SSH)

- web** (http)

How the web works

The **web** (originally called the World Wide Web, thus the “www” in site addresses) is just one of the ways information can be shared over the internet

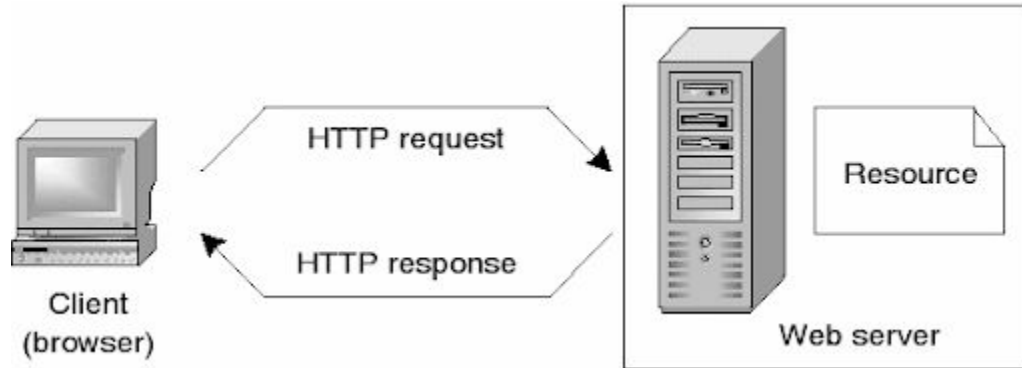
it allows documents to be linked to one another via **hypertext links**—thus forming a huge “**web**” of connected information

The web uses a protocol called **HTTP (HyperText Transfer Protocol)**

How the web works

Web Client

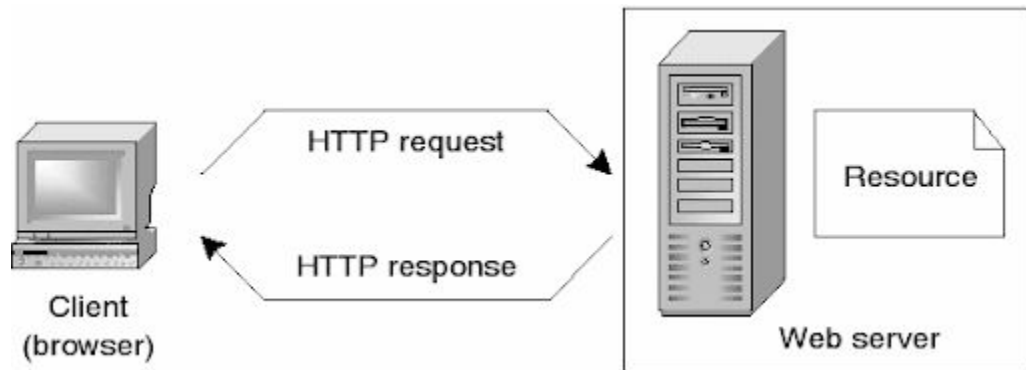
The role of client software is to **make/initiate web requests** and **receive** replies from the web server



How the web works

Web Servers

The role of server software is to **wait for a request for information**, and then **retrieve and send that information back**



How the web works

In order for a computer to be part of the web, it must be running special **web software** that allows it **to handle HyperText Transfer Protocol transactions**

Web Client Software

Browsers - Chrome, Firefox

Curl, Postman



curl://

NGINX

Web Server Software

Apache Web Server, NGINX, IIS



APACHE

HTTP SERVER PROJECT

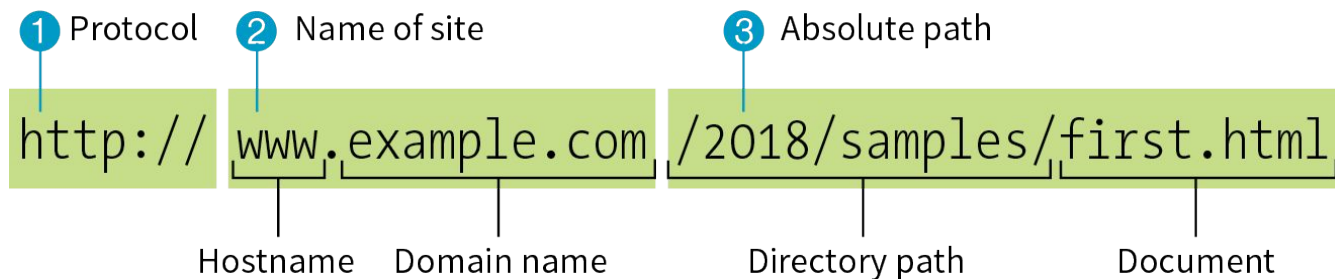
How the web works

How do you locate a particular page?

How the web works

Web Page Addresses (URLS)

Every page and resource on the web has its own special address called a **URL**, which stands for **Uniform Resource Locator**



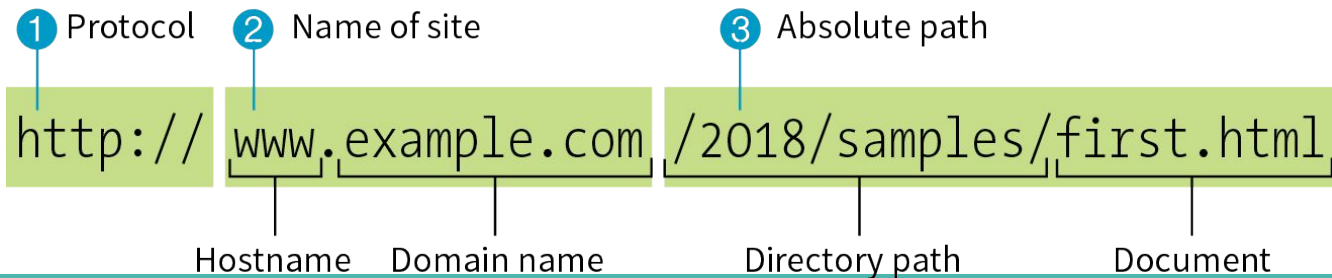
How the web works

The Parts of a URL

http://

let the server know to use HyperText Transfer Protocol

What is the difference between **http** and **https**?



How the web works

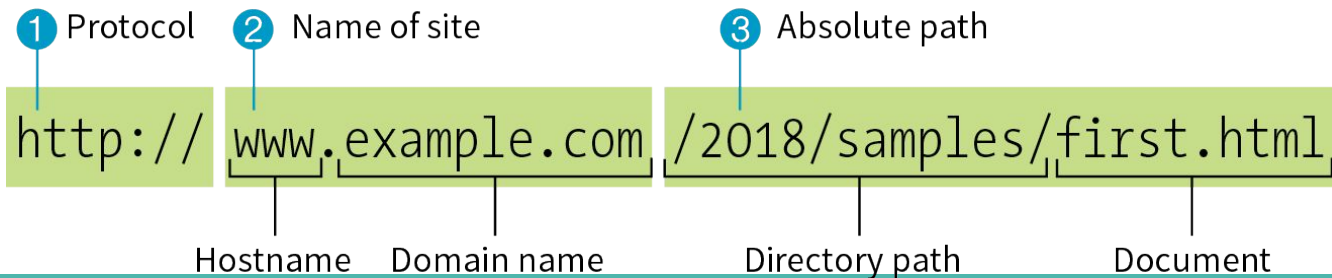
The Parts of a URL

www.example.com

identifies the website by its domain name

the domain name is **example.com**

The **www** part is the particular hostname at that domain



How the web works

IP Address and Domain Name

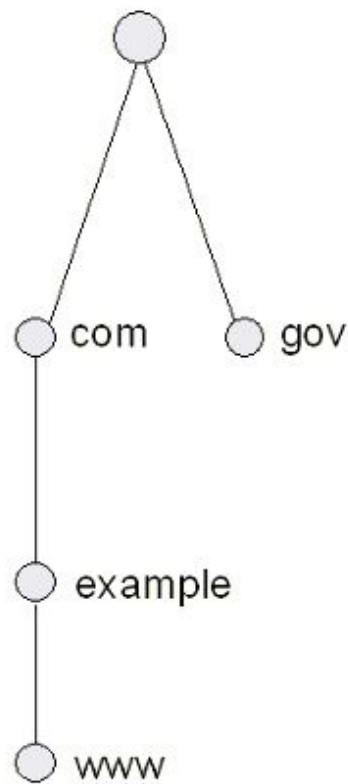
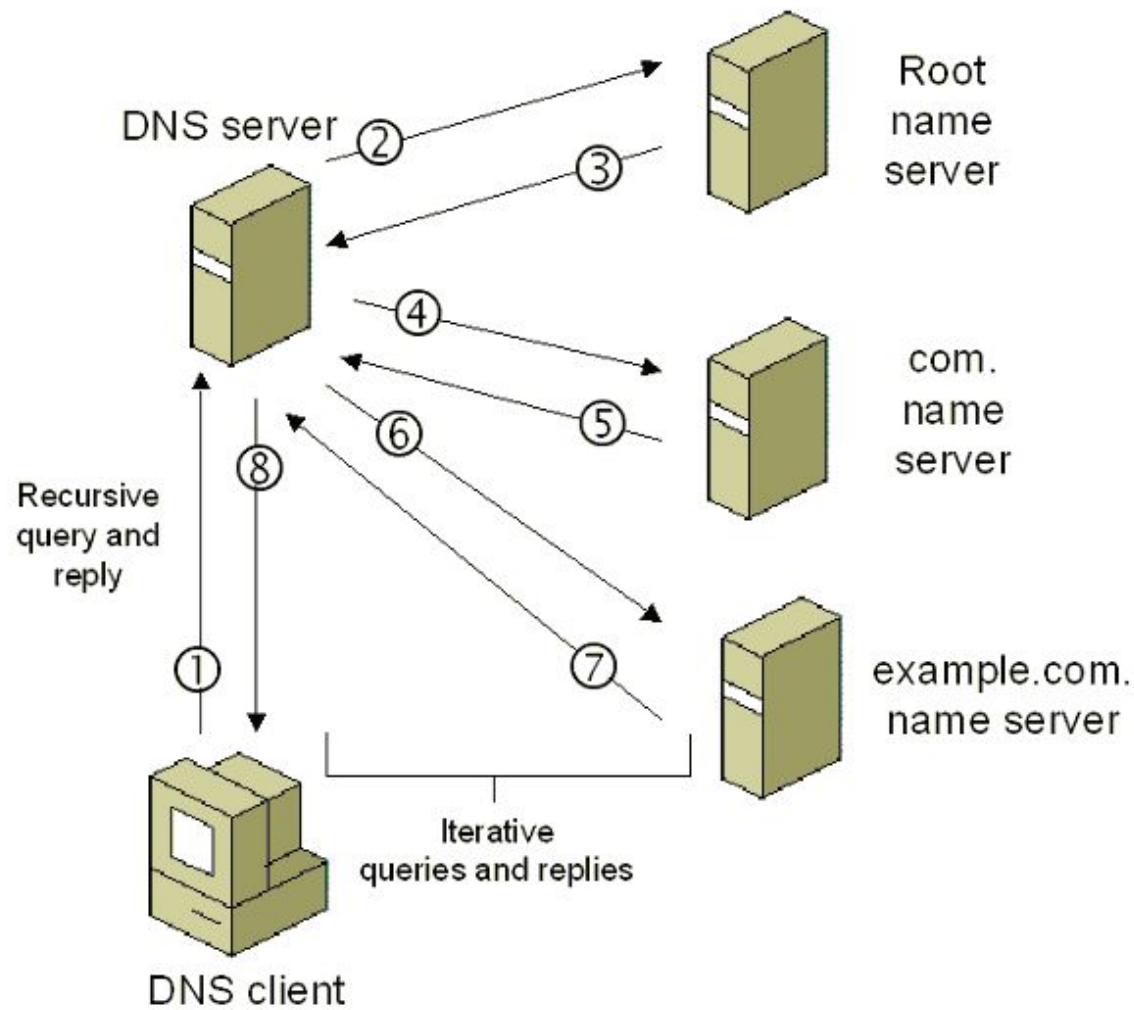
Address that you can use to access computers

DNS Server

Maps a domain names to one or more IP addresses

Next slide shows how the DNS name resolution works

How do you access Google's web server using its IP address?



Exercise

Write the steps and the components involved when you make the a web request using the following url

`http://www.example.com/page1.html`

Consider at least components such as Client (Browser or other command line tools such as curl), Web Server, DNS Server

Browser (Client)

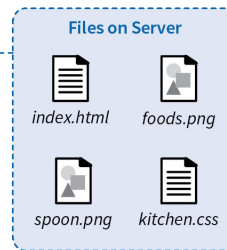
Server

- 1 Type in a URL or click a link in the browser.

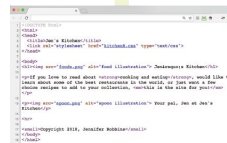
http://jenskitchensite.com

- 2 The browser sends an HTTP request.

HTTP request



HTTP response

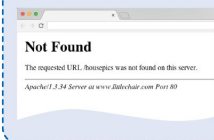


index.html
"I see that you requested a directory, so I'm sending you the default file, index.html. Here you go."

- 4 The browser parses the document. If it has images, style sheets, and scripts, the browser contacts the server again for each resource.

- 3 The server looks for or assembles the file and responds with an HTTP response.

Oops, no file
If the file is not on the server, it returns an error message.



foods.png
kitchen.css
spoon.png

- 5 The page is assembled in the browser window.

Assessment and Evaluation

Assessment	Quantity	Percentage
Lab Exercise	6	15 %
Individual Project	2	25 %
Group Project	1	15 %
Final Exam	1	45 %

All these assessments are required in order to pass the course.

Missing any of them results in NG grade which eventually turns to F if you do not present a valid evidence for missing any of them

Primary References

Frontend

- Learning Web Design - A Beginners Guide to HTML, CSS, Javascript and Web Graphics, 5th edition
- <https://learn.shayhowe.com/html-css/>
- HTML & CSS: Design and Build Websites, 1st edition

Backend

- NestJs Documentation